

How Speakers Work

By: Ava Adiletti

History of the speaker: In 1925, Edward W. Kellogg and Chester W. Rice created the dynamic speaker. Faraday’s law of induction forces the voice coil, a coil of wire that can move axially in a cylindrical gap containing a concentrated magnetic field created by a permanent magnet, to move quickly back and forth when the electrical current from an audio signal passes through it. This coil then attaches to a speaker cone or diaphragm in contact with air (since it is typically conically shaped for durability) to produce sound waves. Several alternative technologies, some of which are in commercial use, can produce sound from an electrical signal in addition to dynamic speakers. (<https://en.wikipedia.org/wiki/Loudspeaker>)

What a speaker is used for: A speaker, also known as a loudspeaker, is a device that converts electrical signals into sound waves, allowing you to hear audio output from various devices like computers, smartphones, TVs, and music players. (<https://www.lenovo.com/us/en/glossary/speakers/?orgRef=https%253A%252F%252Fwww.google.com%252F>)

Impacts and future projections of speakers: The growth of multi-room streaming speakers and the rising demand for music streaming services are expected to propel the global speaker market to \$167.63 billion by 2033 at a compound annual growth rate (CAGR) of 13.8%. The future of audio technology is also anticipated to be significantly influenced by smart speakers and AI-powered audio equipment. (<https://www.businessresearchinsights.com/market-reports/speaker-market-110514>)



<https://www.shutterstock.com/video/search/sound-icon>

Step 1: The Generation of Initial Musical Waveform
When the sound system is powered from zero, the output voltage represents the first musical waveform, which begins by rising. The produced electrical current then starts to bounce back and forth between the positive and negative sides of the voice coil of the sound system.

Step 2: A Magnetic Field is Generated around the Voice Coil
The voice coil has a similar polarity to the magnet in the speaker frame, a magnetic field is created around it. While opposite magnetic fields are drawn to one another, identical magnetic fields repel one another.

Step 3: Air Pressure is Generated
After that, the diaphragm or cone begins to advance, creating air pressure. The audio that humans perceive and hear as sound is created by this air pressure.

Step 4: The Increase in Current Leads to An Increase in the Magnetic Field Strength of the Voice Coil
An increase in electrical current results from the electrical signal voltage rising toward the higher sine wave of the musical signal. This also permits a rise in the voice coil’s magnetic field strength.

Step 5: Further Outward Extension of the Cone
The overall increase allows the cone to project out further.

Step 6: Current Drops to Return the Diaphragm to the Initial Closed Position
The signal begins to decrease after passing through the maximum output. As a result, the diaphragm becomes closer to its initial off position at zero voltage as the electrical current also starts to decrease.

Step 7: The Signal Returns to the Zero Position and the Diaphragm Returns to the Initial Position
The diaphragm returns to its starting position when the signal eventually reaches its zero position, also known as the “zero voltage crossover threshold.”

Step 8: The Signal Starts Its Reverse Process
The signal subsequently turns into a negative voltage and initiates the process in reverse. A magnetic field with reversed polarity is created when electrical current moves from the voice coil’s negative to positive area during this transition.

Step 9: The Action of the Magnetic Field Causes the Diaphragm to Move
The diaphragm starts to move from the front to the back due to an attraction because the voice coil’s magnetic field is the opposite of the speaker’s magnet. This is the reverse of the initial back-to-front route.

Step 10: The Diaphragm Travels in Reverse
The diaphragm moves in reverse as the electrical signal moves along its route. This generates the other half of the sound waves, which were created by air motion.

Step 11: . Stereo Output or System Amplifier Returns to Initial Point. New Cycle Starts
As another electrical signal voltage starts to rise, the stereo output or system amplifier will return to its starting position at zero, where the subsequent signal begins to form. When the audio system starts playing, this will initiate the cycle once more and continue to repeat. Essentially, electrical signals drive speakers, which then convert into mechanical energy that moves air to make sound.

Info from Soundcore.com (8/26/2024)



<https://www.walmart.com/ip/JBL-PartyBox-110-Portable-Party-Bluetooth-Speaker/512669094>



<https://maxpowerus.com/products/mpd425-groove-4-x-2-bluetooth-speaker-with-mic-different-led-light-modes>



<https://www.cnet.com/tech/home-entertainment/best-bluetooth-speaker/>



<https://www.homedepot.com/p/JBL-Charge-5-BT-Speaker-Black-JBLCHARGE5BLKAM/319726892>